# EduNova AI

## Sample Study Guide: Artificial Intelligence, NLP and Deep Learning

A demonstration PDF for testing the document library, AI tutor, smart notes and quiz generator.

*EduNova AI • Version 2.0*

## 1. Artificial Intelligence

Artificial intelligence is the field of building systems that can perform tasks associated with human intelligence, such as reasoning, perception, language understanding and decision making. AI systems combine data, algorithms and computing resources.

## 2. Machine Learning

Machine learning is a subfield of AI. It learns patterns from examples. Supervised learning uses labelled data for classification and regression. Unsupervised learning finds structure in unlabelled data through methods such as clustering.

## 3. Deep Learning

Deep learning uses neural networks with multiple layers. Neurons combine inputs using weights and biases and apply activation functions. Backpropagation calculates gradients, and optimizers update parameters to minimize a loss function.

## 4. Natural Language Processing

NLP enables computers to process human language. Common tasks include tokenization, stop-word removal, stemming, lemmatization, sentiment analysis and named entity recognition. TF-IDF represents term importance across documents.

## 5. Transformer Models

Transformers use self-attention to model relationships between tokens. BERT-style models are useful for language understanding and question answering. BART-style encoder-decoder models can generate summaries. Pretrained models may be used through task-specific pipelines.

## 6. Model Evaluation

Classification evaluation may include accuracy, precision, recall, F1-score and a confusion matrix. Accuracy alone may be misleading for imbalanced data. Models should be evaluated on unseen test data.

## 7. Responsible AI

Responsible AI considers privacy, fairness, transparency, accountability and security. Training data can contain bias, so models should be tested on representative samples and limitations should be communicated. Human review is important for high-impact decisions.

## Quick Revision Checklist

- AI is broader than machine learning.
- Classification predicts categories; regression predicts numerical values.
- CNNs are commonly used for image data.
- Transformers use attention mechanisms.
- Precision and recall complement accuracy.